

Evaluation of the impact of body mass index and mid-upper arm circumference on the quality of cardiopulmonary resuscitation

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Received: 11 April 2025 / Accepted: 28 August 2026

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Abstract

High survival rates in cases of cardiac arrest can only be achieved through high-quality and effective chest compressions. The muscle strength and physical characteristics of the practitioners are among the factors that influence effective chest compressions. This study aims to investigate the relationship between the measurement of mid-upper arm circumference (MUAC), a surrogate anthropometric marker of muscle mass in the body, and the effectiveness of CPR. A total of 142 final-year medical students were selected as participants. The study examined the impact of the participants' physical characteristics, body mass index (BMI), and MUAC measurements on CPR quality. Of the participants, 78 were male, and the rate of achieving appropriate chest compression depth was found to be 99%. MUAC measurements of participants achieving appropriate compression depth ranged between 24 and 28 cm for women and 30–34 cm for men. While most participants with normal BMI achieved the ideal compression rate, no significant correlation was observed between BMI and CPR quality ($n=51$, $p=0.812$). Among women with a MUAC of 24 cm or more, 92.3% ($n=12$) achieved appropriate chest compression depth, whereas this rate was 76.7% ($n=33$) for men. Participants classified as underweight based on BMI had a compression success rate of 8% ($n=7$), compared to 58% ($n=51$) for those with normal BMI. In this study, CPR quality did not differ significantly by MUAC for both genders. Although participants within optimal MUAC ranges achieved adequate compression depth more frequently, these differences were not statistically significant. Participants with a normal BMI were more successful in achieving the ideal compression rate. On the other hand, underweight participants could not maintain adequate chest compression depth.

Keywords Cardiopulmonary resuscitation · Mid-upper arm circumference · Body mass index

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Introduction

International CPR guidelines emphasize the importance of high-quality chest compressions to improve survival rates^{1–3}. Effective CPR can maximize coronary perfusion and the return of spontaneous circulation (ROSC) in patients¹. For effective, uninterrupted, and prolonged chest compressions, it is recommended that practitioners position themselves perpendicular to the patient's body and use their own body weight, thus preventing early muscle fatigue in the practitioner. Besides training and the application of quality CPR, the personal and physical characteristics of the rescuer also influence the effectiveness of CPR. Previous studies have noted that factors such as the rescuer's weight, height, physical fitness, and muscle strength affect the quality of CPR. However, there are limited practitioner-based studies on CPR quality, and existing research is insufficient in this regard^{4–6}.

With technological advancements, various devices have been developed to monitor and improve CPR quality. These include metronomes and feedback devices. CPR feedback devices analyze the quality of resuscitation by providing information on parameters such as compression rate, compression fraction, the ratio of adequate chest compressions, and appropriate chest recoil¹.

While previous studies have demonstrated that higher body weight and BMI are positively associated with achieving adequate chest compression depth, BMI cannot differentiate between muscle and fat mass⁷. Mid-Upper Arm Circumference (MUAC) is a validated, well-established surrogate marker for appendicular skeletal muscle mass in both nutritional and clinical assessments^{8,9}. Therefore, evaluating MUAC as a rapidly obtainable surrogate for muscle mass could offer new insights beyond BMI. Although a recent study has explored arm circumference in relation to CPR¹⁰, comprehensive data combining BMI and MUAC to evaluate their distinct impacts on CPR quality remains scarce.

This study aims to investigate the impact of physiological parameters, including the gender, height, weight, mid-upper arm circumference (MUAC), and Body Mass Index (BMI) of chest compression practitioners, as well as their exercise habits, on the quality of chest compressions.

Methods

Study design

This prospective descriptive observational study was approved by the Non-Interventional Clinical Research Ethics Committee of Hacettepe University, with approval number GO 21/1270, dated December 7, 2021, and decision number 2021/20-28. Consent was obtained from each participant in the study. In the study, training was conducted according to and evaluated according to the American Heart Association's Adult Advance Cardiac Life Support guidelines³.

Between May 1 and August 31, 2022, 6th-year medical students at Hacettepe University Faculty of Medicine who agreed to participate were included in the study conducted at the Department of Emergency Medicine. A post-hoc power analysis indicated that the sample size provided sufficient statistical power to detect moderate effect sizes for the primary outcomes regarding BMI and MUAC. Participants performed uninterrupted chest compressions for 2 min on an adult PRESTAN CPR manikin, used for educational purposes, with a True CPR™ device (Physio-Control®, Redmond, WA, USA), also employed for training purposes.

Data on CPR quality obtained from the True CPR™ device, including chest compression depth, rate, count, compression fraction, and the percentage of compressions achieving proper chest recoil, were recorded in a study form for each volunteer. Chest compression fraction (CCF) was automatically calculated by the True CPR™ device's proprietary software, which continuously logs the exact duration of active chest compressions relative to the total 2-min scenario time.

Each participant's height and weight were measured to calculate their BMI, and their MUAC was also measured.

BMI was calculated using the formula “ $\text{BMI} = \text{kg/m}^2$ ” and classified according to the criteria set by the World Health Organization¹¹. MUAC measurements were taken by a single designated researcher using a standardized measuring tape to minimize inter-rater variability. MUAC measurements were taken while participants were seated, with the forearm flexed at a 90-degree angle and resting on a table, focusing on the midpoint of the humerus bone. Additionally, participants’ age, gender, and exercise habits were recorded in the data form.

Inclusion criteria

To be eligible for the study, participants needed to:

- Be 6th-year students at Hacettepe University Faculty of Medicine.
- Consent to participate in the study.
- Have attended the “Basic Life Support and Advanced Cardiac Life Support” course in the 5th-year Emergency Medicine curriculum.
- Have participated in the “Basic Life Support and Advanced Life Support Algorithms” course and its practical application in the 6th-year Emergency Medicine Internship program.
- Have observed or participated in cardiopulmonary resuscitation in the emergency department at least once.

Exclusion criteria

Exclusion criteria included:

- Failure to provide consent for participation.
- Having physical or chronic conditions that restrict physical activity.

Statistical analysis

All data were analyzed using SPSS Statistics 23 and Jamovi version 2.3.24.0, with a significance level set at $p < 0.05$. Categorical variables were presented as frequency (n) and percentage (%) and compared using χ^2 tests or Fisher’s exact test as appropriate. Normally distributed data were expressed as mean $\pm$ standard deviation (SD), while non-normally distributed data were presented as median (IQR). Comparisons between two groups were made using the Student’s t-test or Mann–Whitney U test, as appropriate.

Results

Between May 1 and August 31, 2022, a total of 142 medical students participating in their emergency medicine internship at Hacettepe University Faculty of Medicine were included in the study. Of the 142 participants, 54.9% (n=78) were male. The median age of the participants was 24 years (IQR: 2). The average height was 173 cm (SD $\pm$ 10.2; minimum 150 cm, maximum 200 cm). The median weight was 70.5 kg (IQR: 26), the median BMI was 23.2 kg/m^2 (IQR: 4.95), and the median MUAC was 30 cm (IQR: 7). The mean compression rate was 104 compressions per minute (range: 79–135), and the compression fraction was calculated as 100%. Since compression fraction was 100% across all groups, it was not included in statistical analyses. The median percentage of adequate chest compressions was 99% (IQR: 8), and the median rate of adequate chest recoil was 88.5% (IQR: 43.5) (Table 1).

Additionally, when the participants’ physical characteristics were examined, those in the group achieving the ideal compression rate had an average height of 174 cm, median weight of 72.5 kg, median BMI of 23.4 kg/m^2 , and median MUAC of 30 cm. None of these physical characteristics yielded a statistically significant difference

Table 1 Physical characteristics of participants.

Characteristic	Mean (SD)/median (IQR)
Gender, Male, n (%)	78 (54.9)
Age, years	24 (2)
Height, cm*	173 (10.2)
Body Weight, kg	70.5 (26)
BMI, kg/m ²	23.2 (4.9)
Mid-Upper Arm Circumference (MUAC), cm	30 (7)
Compression Rate, compressions/min*	104 (9.7)
Adequate Chest Compression Rate (%)	99 (8)
Chest Recoil Adequacy Rate (%)	88.5 (43.5)

*For values with normal distribution, the mean (Standard Deviation) is presented.

For values not following a normal distribution, the median (Interquartile Range) is presented.

Table 2 Physical characteristics and ideal compression rate.

Physical characteristic	Ideal compression rate (100–120 comp/min)		<i>p</i> value
	Yes n = 88 (SD/IQR)	No n = 54 (SD/IQR)	
Age (years)	24 (1.2)	24 (2)	0.740
Height (cm)*	174 (11)	173 (8.6)	0.524
Body Weight (kg)	72.5 (24.5)	68 (22.8)	0.437
BMI (kg/m ²)	23.4 (4.9)	22.5 (5.2)	0.382
MUAC (cm)	30 (7)	28.5 (7.7)	0.384

Comp compression, BMI body mass index.

*Height values with normal distribution are analyzed using Student's *t*-test, other data with.

Mann–Whitney *U* test.

Table 3 Correlation between compression rate and physical characteristics.

Physical characteristic	Correlation coefficient for compression rate	<i>p</i> value
Height	0.056*	0.510
Age	0.081	0.340
Body Weight	0.083	0.327
BMI	0.094	0.266
Mid-Upper Arm Circumference (MUAC)	0.056	0.507
Weekly Exercise Duration	0.092	0.276

*Pearson Correlation Analysis used for height, Spearman's Correlation Analysis for other variables.

Table 4 BMI and ideal compression rate.

BMI Classification	Ideal compression rate (100–120 compressions/min)		Total n (%)
	Yes n = 88 (%)	No n = 54 (%)	
Underweight	7 (8)	4 (7.4)	11 (7.7)
Normal	51 (58)	35 (64.8)	86 (60.6)
Overweight	24 (27.3)	11 (20.4)	35 (24.6)
Obese	6 (6.8)	4 (7.4)	10 (7)
<i>p</i> value	0.812*		

*Cramer's V test; BMI: underweight < 18.5 kg/m², normal 18.5–24.9 kg/m², overweight 25–29.9 kg/m², obese ≥ 30 kg/m².

between those who achieved the ideal compression rate and those who did not (Table 2). No significant correlation was found between compression rate and variables such as height, age, weight, BMI, MUAC, or weekly exercise time (Table 3).

When the impact of BMI on achieving the ideal compression rate was examined, 58% (n = 51) of those who achieved the ideal compression rate had a normal BMI, 27.3% (n = 24) were overweight, 8% (n = 7) were underweight, and 6.8% (n = 6) were obese. No statistically significant difference was found among these groups based on BMI classification (*p* = 0.812) (Table 4).

Table 5 Mid-Upper Arm Circumference (MUAC) and adequate compression depth in women.

MUAC (Women)	Chest compression depth		Total n (%)
	Adequate comp. depth n (%)	Inadequate comp. depth n (%)	
<24 cm	1 (7.7%)	10 (19.6%)	11 (17.2%)
24–28 cm	11 (84.6%)	32 (62.7%)	43 (67.2%)
>28 cm	1 (7.7%)	9 (17.6%)	10 (15.6%)
Total n (%)	13 (%100)	51 (%100)	64 (%100)
<i>p</i> value	0.410*		

*Fisher's exact test used for statistical analysis.

Table 6 Mid-Upper Arm Circumference (MUAC) and adequate compression depth in men.

MUAC (Men)	Chest compression depth		Total n (%)
	Adequate comp. depth n (%)	Adequate comp. depth n (%)	
<30 cm	10 (23.3%)	5 (14.3%)	15 (19.2%)
30–34 cm	23 (53.4%)	23 (65.7%)	46 (59%)
>34 cm	10 (23.3%)	7 (20%)	17 (21.8%)
Total n (%)	43 (%100)	35 (%100)	78 (%100)
<i>p</i> value	0.499		

*Pearson *ki-kare* used for statistical analysis.

For women, the MUAC of those who achieved adequate compression depth (5–6 cm) ranged from 24 to 28 cm. Based on MUAC measurements, female participants were divided into three groups. Among women with MUAC <24 cm, the proportion achieving adequate compression depth was 7.7%, compared to 84.6% in the 24–28 cm group and 7.7% in the >28 cm group. Although the group with MUAC measurements between 24 and 28 cm was the most successful in achieving adequate compression depth, no statistically significant difference was observed among the groups ($p=0.410$) (Table 5).

For men, the MUAC of those who achieved adequate compression depth (5–6 cm) ranged from 30 to 34 cm. Among men with MUAC <30 cm, the proportion achieving adequate compression depth was 23.3%, compared to 53.5% in the 30–34 cm group and 23.3% in the >34 cm group. Although the group with MUAC measurements between 30 and 34 cm was the most successful in achieving adequate compression depth, no statistically significant difference was observed among the groups ($p=0.499$) (Table 6).

Discussion

The performance of effective chest compressions has been recognized as a key determinant of successful outcomes in cardiac arrest for over 40 years¹². Both the American Heart Association (AHA) and the European Resuscitation Council (ERC) emphasize the importance of high-quality chest compressions as the cornerstone of resuscitation care^{1,2}. In this study, the effects of BMI and MUAC on chest compression quality were investigated within the framework of the AHA's core criteria.

It has been shown in earlier research that the mid-upper arm circumference (MUAC) is a reliable indicator of muscle mass in the body^{8,13}. Recent studies have demonstrated a positive correlation between arm muscle thickness and overall muscle mass^{14,15}. Since MUAC is practical and does not require specialized equipment, it is highly accessible. Aygun et al.¹⁰ reported significant positive correlations between maximum biceps circumference and both compression quality and overall CPR quality, whereas no significant association between MUAC and compression depth was observed in our cohort; this discrepancy may reflect differences in anthropometric measurement methods and participant characteristics. However, few studies have examined the specific effect of MUAC on CPR performance, making this study one of the rarely to evaluate MUAC in conjunction with BMI as a determinant of CPR quality.

In our study, the average MUAC was found to be 26 cm in women and 32.5 cm in men. The highest proportion of adequate compression depth was observed in participants with MUAC values of 24–28 cm among women and 30–34 cm among men; however, these differences were not statistically significant. Although participants within these specific MUAC ranges achieved higher rates of adequate depth, the study found no significant correlation between MUAC and the ability to achieve the ideal compression rate or adequate compression depth for either gender ($p=0.410$ for women, $p=0.499$ for men). Conversely, participants who achieved adequate chest recoil had specific MUAC ranges, with women being more successful in achieving complete chest recoil during CPR compared to men.

When examining CPR quality indicators by gender, this study found that both genders had similar compression rates. However, the percentage of compressions performed at an adequate depth (5–6 cm) was 96% for women and 100% for men, with men achieving higher rates of adequate compression depth. Previous studies by Ok et al., Kim et al., Zhang et al., Mayrand et al., and Russo et al. have similarly reported better compression depth performance in men^{6,16–19}. According to these findings, female participants tend to perform faster but shallower chest compressions compared to their male counterparts. Consistent with the literature, our study also found that male participants achieved higher rates of adequate compression depth, likely due to their greater muscle mass. Muscle size and overall CPR performance can be influenced by various potential confounding factors, including training history, sex, age, and nutritional status. Although we focused on MUAC, it is essential to recognize that repeated CPR training may build muscle memory and technique that can compensate for differences in baseline muscle mass.

When examining the relationship between BMI and adequate compression depth, a weak positive correlation was found. Mayrand et al. reported a positive correlation between chest compression depth and increases in weight and height¹⁹. On the other hand, Ock et al. found no significant relationship between adequate compression depth and variables such as gender, height, or weight²⁰. Krikščionaitienė et al. found that participants classified as underweight were less likely to perform adequate chest compressions²¹. Sánchez et al. noted that individuals with a BMI ≥ 23 kg/m² maintained an adequate compression depth for 2 min, while those with a BMI < 23 kg/m² performed effectively only 80% of the time in the first minute, dropping to 30% by the second minute²². Similarly, Wang et al. found that individuals with above-average BMI (mean BMI 21.6 ± 2.3 kg/m²) performed better in CPR quality parameters²³. Oh et al. showed that rescuers with higher BMI were more successful in achieving adequate compression depth²⁴. Similarly, Eyupoglu et al.⁷ found that participants who performed optimal CPR had significantly higher BMI values than those with suboptimal CPR, while BMI was not significantly associated with CPR performance in our study, possibly due to differences in study populations, training backgrounds, and CPR assessment protocols.

The general conclusion from previous studies is that anthropometric variables like height and weight can influence chest compression quality to some extent^{24–26}. Higher BMI increases the likelihood of success in achieving adequate compression depth, but it may inversely affect chest recoil rates. Additionally, the greater muscle mass in men contributes to higher rates of adequate compression. Conversely, individuals with lower BMI face greater challenges in achieving adequate chest compression depth on CPR manikins^{27,28}. Although previous studies have suggested that gender and BMI may influence CPR performance, our findings did not demonstrate statistically significant associations between BMI or MUAC and the CPR quality parameters evaluated.

Limitations

This study has several limitations. First, the single-center design and the specific cohort of 6th-year medical students limit generalizability. By the time students reach the sixth year, they have already received BLS training in their fifth year; this prior exposure and repeated training likely homogenized their skill levels, which could have masked the isolated impact of physical characteristics like MUAC on CPR quality. Although quantitative CPR quality data from their 5th-year training were not recorded, all participants had successfully passed their

practical and theoretical internship examinations, ensuring a uniform baseline level of competence. Second, the 2-min CPR sessions on a manikin cannot fully evaluate the "real" quality of CPR at the scene, as a brief simulation fails to replicate the physical fatigue and dynamic challenges of a prolonged clinical cardiac arrest scenario. Furthermore, BMI cannot differentiate between muscle mass and fat mass, and the relatively narrow range of MUAC within our specific, young cohort of medical students may have limited our ability to detect statistically significant differences.

Conclusion

In conclusion, this study found no statistically significant difference in CPR quality based on MUAC for both genders. While an increase in MUAC showed a tendency towards better compression depth, it did not reach statistical significance. In contrast, higher MUAC was associated with lower rates of adequate chest recoil. Furthermore, CPR practitioners with a normal BMI were more successful in achieving the ideal compression rate, while underweight practitioners struggled to perform adequate chest compressions.

Acknowledgements None.

Author contributions S.Ö and M.A.K designed the study. S.Ö, G. A, B. C. B and M.A.K analysed and interpreted the data and was the main contributor in writing the manuscript. S.Ö, B.E., M.A.K and E. Ö. İ proved the statistical analysis and the correctness of the given information about the project details. S. Ö, I.A and M.A. K was a major contributor in writing the manuscript and developing the research question. All authors read and approved the final manuscript.

Funding The authors declared that this study has received no financial support.

Data availability The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests No conflict of interest was declared by the authors.

Ethics approval This study was approved by the Non-Interventional Clinical Research Ethics Committee of Hacettepe University, with approval number GO 21/1270, dated December 7, 2021, and decision number 2021/20-28.

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